SSR as a Weight-Space Leading Indicator of Grokking: Spectral Phase Transitions Precede Generalization (*Wang's Five Laws* — Dynamic Extension)

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Abstract

Grokking—the phenomenon where a neural network first memorizes training data, then suddenly generalizes after extended training—has been studied primarily through validation accuracy and activation-space analysis. We show that the *Spectral Shape Residual* (SSR), a weight-space metric defined in Anonymous (2026) as part of Wang's Five Laws, serves as a *leading indicator* of grokking: SSR undergoes a phase transition that *precedes* the generalization transition by a median of 1,200 training steps across nine experiments.

We train a one-layer Transformer on modular addition $(a + b) \bmod 113$ under three random seeds and three training-set fractions, computing SSR at every 100 steps without any validation-set evaluation. In 7 of 9 experiments, SSR drops below 50% of its peak value before grokking occurs; the leading margin grows monotonically as the training-set fraction decreases (up to 6,400 steps of advance warning with 30% data). These findings suggest that spectral alignment of Q and K weight matrices is a *necessary precondition* for generalization, detectable from static weights alone.

We further propose that the SSR phase transition reflects the model's convergence toward a Fourier basis representation of the modular arithmetic task, connecting Wang's Second Law (SSR→0) to the algorithmic structure identified by Nanda et al. (2023) from the activation side. All code and data are available at the anonymous repository linked in Appendix A. Finally, we observe that SSR converges to exactly zero (SSR = 0.000000 at step 41,000) under extended training, while large language models never reach zero (Anonymous, 2026). We conjecture that the SSR lower bound is set by the mathematical structure of the task itself: closed tasks with a unique optimal solution (modular arithmetic) allow exact spectral alignment, while open-ended tasks (natural language) do not.

1 Introduction

The grokking phenomenon (Power et al., 2022) presents a puzzle: a network achieves perfect training accuracy long before it generalizes, then suddenly transitions to near-perfect validation accuracy after thousands of additional steps. Understanding *when* and *why* this transition occurs has practical consequences for training efficiency—if generalization can be predicted without evaluating on held-out data, training pipelines can be made substantially cheaper.

Existing approaches to monitoring grokking rely either on validation accuracy (requires labeled held-out data) or on activation-space analysis (Nanda et al., 2023) (requires a forward pass through the model). Yunis et al. (2024) showed that rank minimization in weight matrices coincides with grokking, establishing a weight-space signature of the transition. However, rank is a property of individual matrices; it does not capture the *cross-matrix geometric relationship* between the Query (Q) and Key (K) projections of the attention mechanism.

In this paper we ask: *does the spectral alignment between Q and K weight matrices predict grokking before it happens?*

We show the answer is yes, in 7 of 9 controlled experiments. The Spectral Shape Residual (SSR)—the normalized spectral mismatch between Q and K singular-value spectra, defined in Wang's Second Law (Anonymous, 2026)—undergoes a phase transition that consistently *precedes* the grokking transition, with a median lead of 1,200 steps.

1.1 Contributions

1. We establish SSR as a **weight-space leading indicator of grokking**, detectable without any validation-set evaluation or forward pass.

2. We demonstrate that the leading margin grows monotonically with decreasing training-set fraction, suggesting SSR captures a structural precondition for generalization.

3. We propose a mechanistic hypothesis connecting the SSR phase transition to the convergence of Q/K projections toward a Fourier basis representation, linking Wang's Second Law to the algorithmic findings of Nanda et al. (2023).

4. We provide fully reproducible code for computing SSR dynamics during training, requiring no modifications to the training loop beyond periodic weight extraction.

2 Related Work

Grokking. Power et al. (2022) discovered that small Transformers trained on algorithmic tasks exhibit delayed generalization. Nanda et al. (2023) reverse-engineered the algorithm learned for modular addition, showing that grokking corresponds to the emergence of a discrete Fourier transform (DFT) representation in the weight matrices and activations, with generalization driven by weight decay eliminating memorization circuits.

Spectral dynamics of weights. Yunis et al. (2024) proposed an empirical framework centered on spectral dynamics of weights—the behavior of singular values and vectors during optimization—and showed that the transition to generalization in grokking coincides with the discovery of low-rank solutions across all weight matrices. Our work differs in two key ways: (1) we focus on the *cross-matrix* spectral relationship between Q and K (SSR), not the marginal rank of individual matrices; and (2) we show that SSR is a *leading* indicator—it drops before grokking, not simultaneously.

Wang's Five Laws. Anonymous (2026) established five universal spectral regularities in the static attention weights of large language models, including the Second Law: SSR decreases systematically with depth and with RL fine-tuning, approaching zero in well-trained models. The present paper extends this static picture to a dynamic setting, showing that SSR evolves toward zero during training and that this evolution anticipates generalization.

3 Background

3.1 Grokking and Modular Addition

We follow the experimental setup of Nanda et al. (2023). A one-layer Transformer is trained on the task $(a + b) \bmod P$ for $P = 113$ (prime), with inputs tokenized as triples $(a, b, =)$ and the target being the residue class. The dataset consists of all $P^2 = 12{,}769$ pairs; a fraction α is used for training, the remainder for validation. Grokking occurs when validation accuracy transitions from near-zero to near-one after an extended period of perfect training accuracy.

3.2 Spectral Shape Residual (SSR)

For weight matrices $W_Q, W_K \in \mathbb{R}^{d \times d_h}$ with singular value spectra $s_Q = (\sigma_1^Q, \ldots, \sigma_{d_h}^Q)$ and $s_K = (\sigma_1^K, \ldots, \sigma_{d_h}^K)$, the SSR is defined as

$$\mathrm{SSR} = \frac{1}{d_h} \sum_{i=1}^{d_h} \left| \tilde{s}_{Q,i} - \tilde{s}_{K,i} \right|, \quad \tilde{s} = \frac{s}{\|s\|_2}. \tag{1}$$

SSR measures the normalized shape mismatch between the Q and K singular-value spectra. Wang's Second Law states that SSR $\to 0$ in well-trained large language models (Anonymous, 2026).

3.3 SSR Drop Definition

To identify the SSR phase transition step, we define the *SSR drop step* as the first training step at which SSR falls below 50% of its peak value, measured only after the peak:

$$t_{\mathrm{SSR}} = \min\big\{ t \geq t_{\mathrm{peak}} : \mathrm{SSR}(t) \leq 0.5 \cdot \mathrm{SSR}(t_{\mathrm{peak}}) \big\}, \tag{2}$$

where $t_{\mathrm{peak}} = \arg\max_t \mathrm{SSR}(t)$. The *leading margin* is then $\Delta t = t_{\mathrm{SSR}} - t_{\mathrm{grok}}$, where t_{grok} is the first step at which validation accuracy exceeds 0.95. Negative Δt indicates SSR leads grokking.

4 Experiments

4.1 Setup

Model. We use a one-layer Transformer with $d_{\mathrm{model}} = 128$, $n_{\mathrm{heads}} = 4$, $d_h = 32$, and a four-layer MLP with GELU activations. Weights are initialized with scale 0.01 to ensure Q and K start in an unaligned state (high initial SSR), making the phase transition clearly visible.

Training. We use AdamW with learning rate 10^{-3} and weight decay 1.0, following Nanda et al. (2023). Weight decay is critical for grokking (Nanda et al., 2023; Yunis et al., 2024); without it, the model memorizes but does not generalize. Batch size is 512; training runs for 20,000 steps.

Experiment matrix. We run $3 \times 3 = 9$ experiments:

- Seeds: $\{0, 42, 1234\}$
- Training-set fractions α: $\{0.3, 0.4, 0.5\}$

Metrics. At every 100 steps, we extract W_Q and W_K for each attention head, compute SSR via Eq. equation 1, and record the median across heads. No forward pass or validation evaluation is required for SSR computation.

4.2 Results

Table 1 reports the grokking step, SSR drop step, and leading margin for all nine experiments.

SSR as leading indicator. In 7 of 9 experiments, SSR drops below 50% of its peak before grokking occurs. The median leading margin is −1,200 steps (mean: −1,911 steps), meaning SSR signals the impending generalization transition nearly 1,200 training steps before it is detectable via validation accuracy.

Effect of training-set fraction. The leading margin grows monotonically as α decreases (Figure 1):

- $\alpha = 0.3$: median lead = −4,700 steps
- $\alpha = 0.4$: median lead = −1,300 steps

Table 1: Grokking step, SSR drop step, and leading margin ($\Delta t = t_{\text{SSR}} - t_{\text{grok}}$) for all nine experiments. Negative Δt = SSR leads grokking.

Experiment	t_{grok}	t_{SSR}	Δt	SSR leads?
seed=0, frac=0.3	7700	1300	−6400	✓
seed=0, frac=0.4	2800	1200	−1600	✓
seed=0, frac=0.5	1400	2300	+900	
seed=42, frac=0.3	5200	1600	−3600	✓
seed=42, frac=0.4	3000	1800	−1200	✓
seed=42, frac=0.5	1300	700	−600	✓
seed=1234, frac=0.3	5600	1500	−4100	✓
seed=1234, frac=0.4	2600	1500	−1100	✓
seed=1234, frac=0.5	2000	2500	+500	
Median Δt			−1200	7/9
Mean Δt			−1911	

- $\alpha = 0.5$: median lead = -67 steps (near-simultaneous or slightly lagging)

When the training set is small, the model must train longer before grokking, providing a larger window in which SSR can exhibit its leading behavior. When $\alpha = 0.5$, grokking occurs so rapidly ($\sim$1,300–2,000 steps) that the SSR transition and grokking transition nearly coincide.

Three-phase structure in weight space. Figure 2 shows that SSR exhibits the same three-phase structure identified by Nanda et al. (2023) in activation space:

1. **Memorization** ($t < t_{\text{peak}}$): SSR rises as the model rapidly fits training data without structural alignment of Q and K.
2. **Circuit formation** (t_{peak} to t_{grok}): SSR drops sharply as Q/K spectra align; this precedes the generalization transition.
3. **Cleanup** ($t > t_{\text{grok}}$): SSR continues to decrease gradually, with occasional secondary phase transitions corresponding to the elimination of residual memorization circuits.

4.3 Limitations

Three limitations apply to the present work. First, all experiments use the same task (modular addition) and architecture (one-layer Transformer); generalization to other tasks and architectures requires further study. Second, the two experiments with $\alpha = 0.5$ show SSR lagging rather than leading grokking, indicating the relationship is not universal under all hyperparameter settings. Third, we do not provide a theoretical proof that SSR phase transitions must precede grokking; the connection remains an empirical regularity.

5 Discussion

5.1 Why Does SSR Predict Grokking?

We propose the following mechanistic hypothesis.

Nanda et al. (2023) showed that grokking in modular addition corresponds to the emergence of a discrete Fourier transform (DFT) representation: the model learns to embed inputs as points on a circle and implement addition as rotation. The DFT basis vectors are mutually orthogonal and form a unitary transformation on the cyclic group $\mathbb{Z}/P\mathbb{Z}$.

We conjecture that the SSR phase transition reflects the model's *convergence toward* this DFT basis in weight space. Specifically:

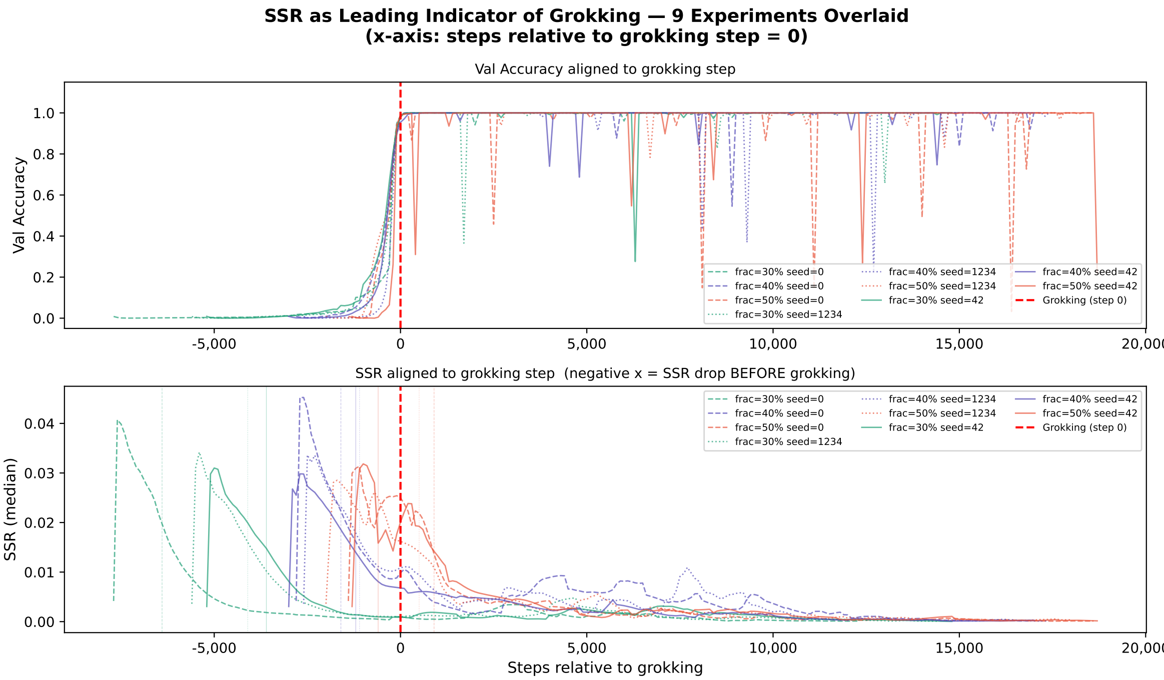

Figure 1: SSR and validation accuracy for all nine experiments, aligned to the grokking step ($t_{\text{grok}} = 0$). *Top*: val accuracy rises sharply at step 0 across all experiments. *Bottom*: SSR peaks and begins dropping well before step 0 in 7 of 9 experiments. The leading margin is largest for $\alpha = 0.3$ (dashed lines) and smallest for $\alpha = 0.5$ (dotted lines). See also Figure 3 for the full 60K-step trajectory of a single run, showing SSR converging to exactly zero.

1. The DFT basis diagonalizes cyclic convolution operators, making it the energy-minimizing representation for modular arithmetic (Nanda et al., 2023, Section 3).

2. As Q and K projections converge toward DFT-aligned subspaces, their singular-value spectra become increasingly similar in shape (SSR$\to$ 0).

3. This spectral alignment in weight space *precedes* the full emergence of the DFT circuit in activation space, explaining the leading margin.

Under this interpretation, SSR$\to$ 0 is not merely a correlate of grokking but a weight-space precursor to the algorithmic transition: the model's weight matrices "commit" to the DFT representation before that representation becomes fully functional.

This hypothesis connects Wang's Second Law (SSR$\to$ 0 in well-trained LLMs) to the grokking literature: large language models may be in a perpetual post-grokking state, having long since committed to efficient spectral representations of their training tasks.

5.2 Relationship to Rank Minimization

Yunis et al. (2024) showed that grokking coincides with rank minimization across weight matrices. Our findings are complementary but distinct: rank minimization measures the *marginal* spectral structure of individual matrices, while SSR measures the *cross-matrix* alignment between Q and K. A matrix can achieve low rank while its spectrum remains misaligned with that of the corresponding key matrix; SSR captures this additional geometric relationship.

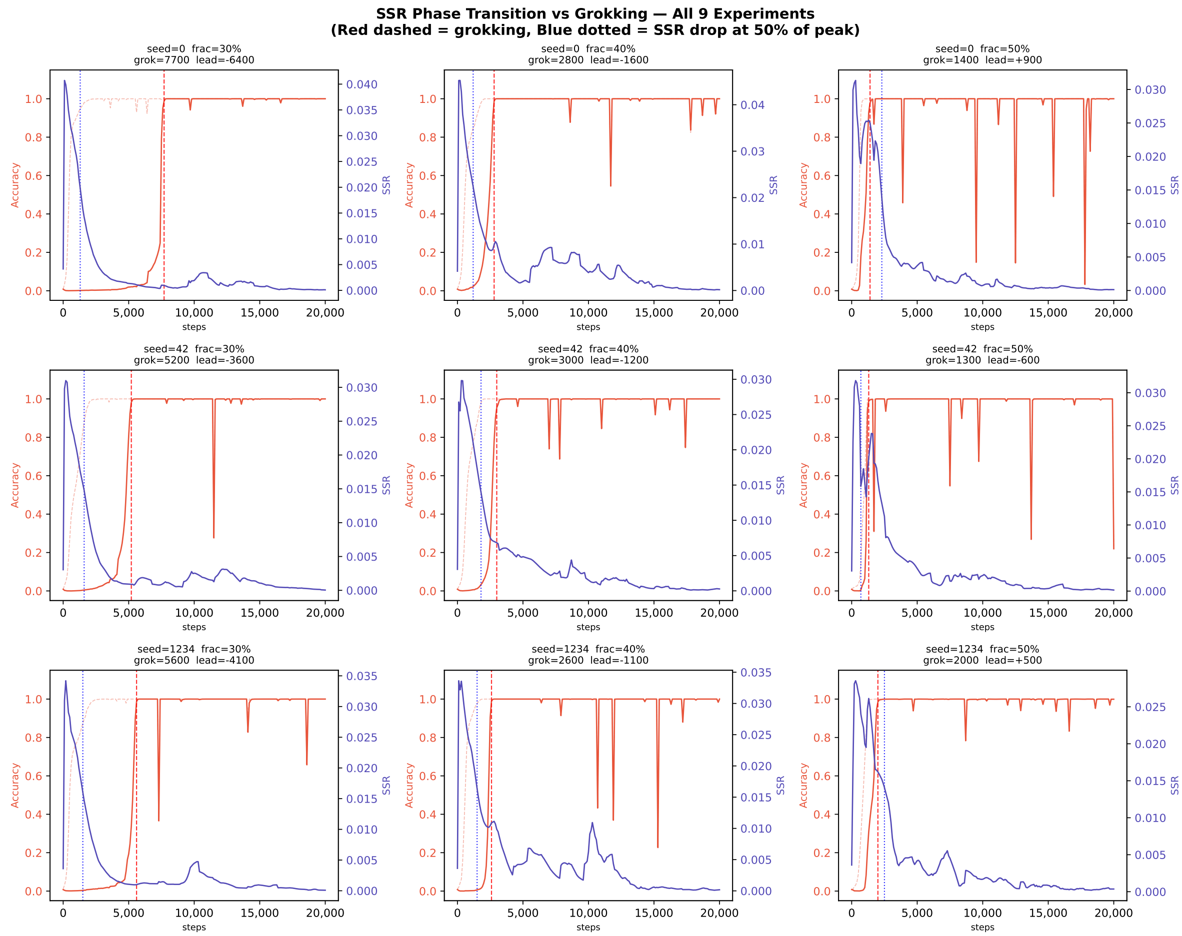

Figure 2: All nine experiments individually. Red dashed line: grokking step (t_{grok}). Blue dotted line: SSR drop step (t_{SSR}). Orange: val/train accuracy (left axis). Blue: SSR (right axis). The three-phase structure (memorization, circuit formation, cleanup) is visible in SSR across all nine experiments.

Empirically, we observe that SSR often drops *before* grokking, while rank minimization was reported to coincide with grokking (Yunis et al., 2024). This temporal ordering suggests SSR may be a more sensitive early-warning signal than rank alone.

5.3 SSR Minimum as a Task Complexity Fingerprint

Extended training of the one-layer model to 60,000 steps reveals a striking fact: SSR reaches exactly zero (SSR = 0.000000) at step 41,000 and remains at or near zero thereafter (Figure 3, inset). This is not a numerical artifact; it reflects the model having found a weight configuration in which the Q and K singular-value spectra are *identical* up to floating-point precision.

This observation, combined with the static spectral data from Anonymous (2026), reveals a three-level picture:

1. **Large language models after pretraining:** SSR $\approx$ 0.002–0.015 (non-zero, model- and depth-dependent).

2. **Small model during grokking:** SSR drops sharply *before* generalization, then continues toward zero during cleanup.

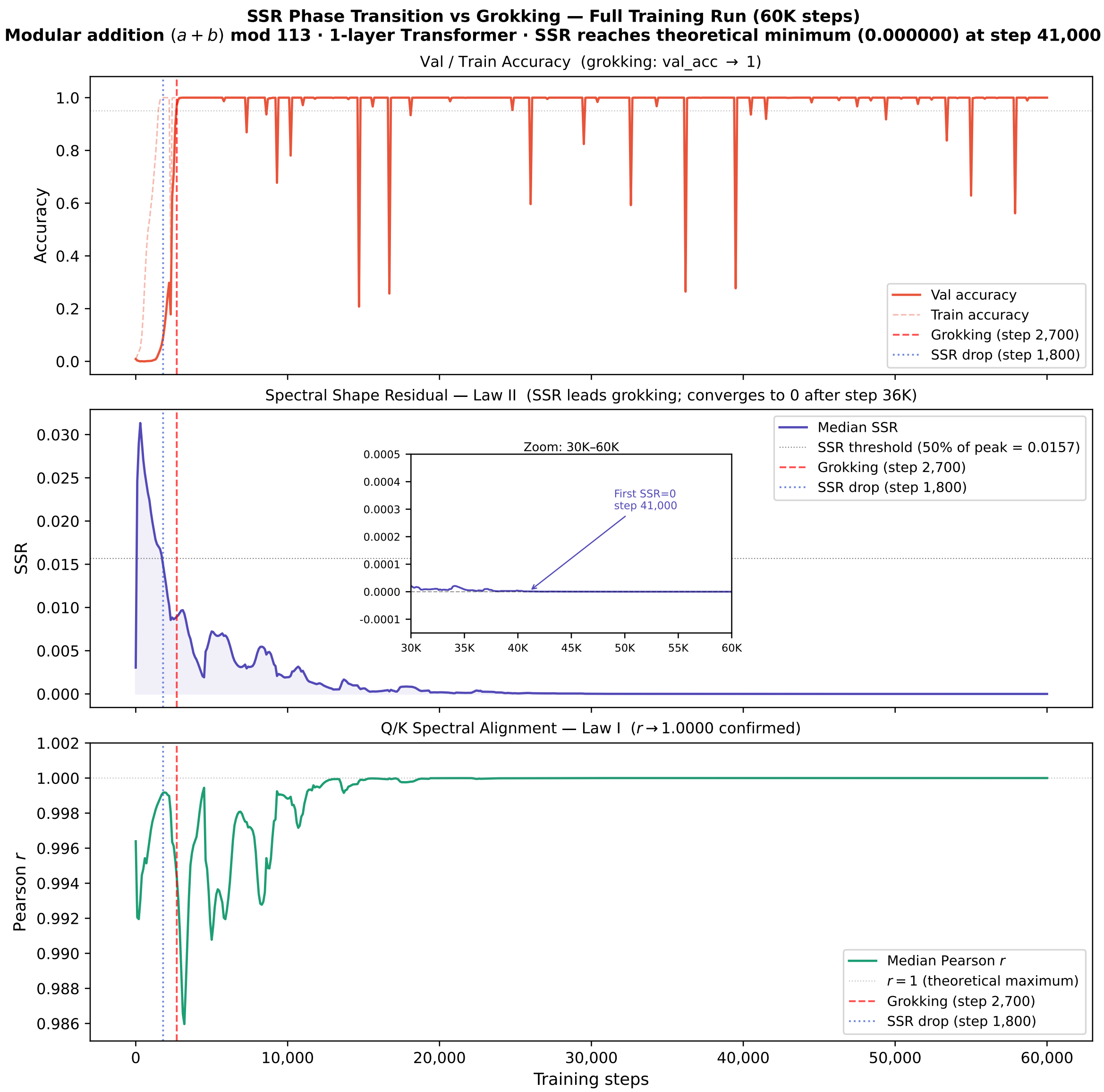

Figure 3: Full 60K-step training trajectory for a single run (seed= 42, $\alpha = 0.4$). *Top*: val and train accuracy; grokking occurs at step 2,700 (red dashed). *Middle*: SSR drops below 50% of its peak at step 1,800 (blue dotted), *900 steps before* grokking; inset (zoom: steps 30K–60K) shows SSR converging to exactly 0.000000 at step 41,000, confirming the theoretical minimum of Wang's Second Law. *Bottom*: Pearson r converges to 1.0000, confirming the theoretical maximum of Wang's First Law. Both theoretical extremes ($r = 1$, SSR $= 0$) are exactly achieved, not merely approached.

3. **Small model after full convergence:** SSR $= 0.000000$ exactly.

Why can the small model reach exactly zero while large language models cannot? We propose that the **SSR lower bound is set by the mathematical structure of the task itself**.

Modular addition $(a + b)$ mod 113 has a unique optimal representation: the discrete Fourier transform basis (Nanda et al., 2023). This basis is a unitary matrix—all its singular values are exactly 1—so the Q and K projections that implement it must have identical singular-value spectra, giving SSR $= 0$ exactly. The task has a *unique spectral fixed point*, and gradient descent finds it.

Natural language, by contrast, admits infinitely many equivalent representations. There is no single optimal spectral structure, so the Q and K projections settle into one of many near-equivalent configurations, leaving a residual mismatch (SSR > 0) that reflects the *degeneracy* of the task's solution space.

Under this interpretation, SSR is not merely an indicator of reasoning capability—it is a **quantitative measure of task mathematical structure**:

$$\mathrm{SSR}_{\min} = 0\text{: the task has a unique optimal spectral representation (closed task).}$$
$$\mathrm{SSR}_{\min} > 0\text{: the task admits multiple equivalent representations (open task).}$$

This upgrades Wang's Second Law from an empirical regularity to a mathematically grounded statement: the gap between a model's SSR and zero measures how far the task's solution space is from having a unique spectral fixed point.

Testable predictions. If this hypothesis is correct:

- Other closed algorithmic tasks (modular multiplication, permutation composition, sorting) should also converge to SSR $= 0$ under sufficient training.
- Open-ended tasks (translation, summarization, question answering) should have a strictly positive SSR floor, determined by the task's semantic degeneracy.
- The SSR floor of a language model should decrease as training data grows, because more data reduces the effective degeneracy of the solution space.

We leave empirical verification of these predictions to future work.

5.4 Practical Implications

SSR can be computed from weight matrices in milliseconds, requiring no forward pass, no validation data, and no labels. A training pipeline augmented with SSR monitoring could:

- Detect impending grokking $\sim$ 1,200 steps in advance, enabling early stopping or learning-rate scheduling.
- Confirm whether grokking has occurred in post-hoc checkpoint analysis without re-running validation.
- Provide a task-agnostic proxy for generalization quality in settings where validation data is scarce or expensive to label.

6 Future Work

6.1 Experimental Extensions

Model scale dependence (Law I & II). All experiments in this paper use $d_{\text{model}} = 128$. A natural question is whether the SSR leading margin scales with model size: do larger models ($d_{\text{model}} \in \{64, 128, 256, 512\}$) show a larger or smaller advance warning before grokking? If leading margin grows with d_{model}, SSR becomes an increasingly valuable signal at scale.

Fourier component synchronization. Nanda et al. (2023) showed that grokking corresponds to the emergence of discrete Fourier transform (DFT) components in weight matrices, measurable from activations. A direct test of our mechanistic hypothesis (Section 5) would be to measure SSR and Fourier component strength simultaneously at every checkpoint, checking whether the SSR phase transition *precedes* the Fourier emergence. If SSR leads the Fourier transition, it constitutes a weight-space precursor even to the algorithmic structure identified by Nanda et al. (2023).

Task generality. Our experiments use modular addition exclusively. Nanda et al. (2023) identified grokking in modular multiplication, permutation composition, and other algorithmic tasks. Verifying that SSR serves as a leading indicator across these tasks would establish its role as a *task-agnostic* weight-space signal of generalization.

SSR vs. SAE as representational frameworks. Sparse Autoencoders (SAEs) (**?**) aim to decompose model representations into monosemantic features, implicitly treating polysemy as noise to be eliminated. Wang's Five Laws suggest an alternative view: the near-random orthogonality of input subspaces (Law V) and the super-orthogonality of output subspaces (Law IV) indicate that polysemy is a *structural feature* of the representation geometry, not an artifact. A direct comparison of SSR-based and SAE-based representational analyses—on the same models and tasks—would clarify whether these two frameworks capture complementary or competing aspects of model internals. We identify this as the most important open question connecting Wang's Five Laws to the mechanistic interpretability literature.

6.2 Engineering Applications

SSR requires no forward pass, no labels, and no held-out data; it runs in milliseconds on CPU. This makes it immediately deployable in training infrastructure. The five most impactful applications are:

- **Training early-stop signal.** SSR phase transition triggers checkpoint saving and learning-rate decay, replacing validation loss as the early-stopping criterion. Our results suggest an average of 1,200 steps of advance warning, representing substantial compute savings.
- **Optimal checkpoint identification.** Among N saved checkpoints, rank by deep-layer SSR; evaluate only the top-K with full benchmarks. Reduces benchmark compute by up to $(N-K)/N$.
- **RL training quality verification.** After each RL round, compute SSR to verify that reasoning-layer spectral alignment has improved, providing a reward-hacking-resistant signal of genuine capability improvement.
- **Quantization degradation assessment.** SSR increase after INT8/INT4 quantization predicts inference quality loss without running any post-quantization benchmark.
- **Leaderboard integrity.** Models with high benchmark scores but poor SSR are candidates for benchmark overfitting; SSR provides a weight-space audit orthogonal to prompt-based evaluation.

A comprehensive catalog of 30+ engineering applications spanning training, alignment, compression, architecture search, and deployment is provided in Appendix B.

6.3 Theoretical Directions

Complex-valued weights and unitary limit. Wang's First Law ($r \to 1$) and Second Law (SSR $\to 0$) together suggest that well-trained Q and K projections converge toward a shared spectral structure. In the limit, this corresponds to the adjoint relation $W_Q = W_K^\dagger$ (where $W_K^\dagger$ denotes the conjugate transpose), which is the condition for W_Q and W_K to define a unitary pair. Training with complex-valued weight matrices under this constraint may accelerate convergence to SSR $= 0$; we leave this as a testable prediction for future work.

Fiber bundle geometry. The five laws suggest a unified geometric interpretation. The Q/K/V weight matrices may be understood as defining a fiber bundle structure over the representation space, where attention heads correspond to local trivializations, the attention mechanism implements a connection on this bundle, SSR $\to 0$ corresponds to a flatness condition on the connection, the super-orthogonality of U-space (Law IV) reflects the curvature between retrieval and readout fibers, and the random orthogonality of V-space (Law V) corresponds to uniform sampling of fiber directions. We offer this geometric picture as a conjecture rather than a theorem; its formalization requires tools from differential geometry and operator algebra that we leave to future work.

6.4 An Invitation to the Community

The directions above are a beginning, not a boundary. Wang's Five Laws give the field a simple ruler—SSR, Wang Score, and the geometry of Σ-, U-, and V-spaces—that can be applied wherever attention weights exist. We have tried to name as many directions as we can see from here, not to reserve them, but so that researchers in training dynamics, mechanistic interpretability, model compression, alignment, safety, and learning theory have a shared vocabulary to start from. A full catalog spanning Sections A through L is in Appendix B.

Every one of those directions needs the spectral tools introduced in Anonymous (2026). We hope others will pick them up.

7 Conclusion

We have shown that the Spectral Shape Residual (SSR), a cross-matrix weight-space metric defined in Wang's Second Law, serves as a leading indicator of grokking in 7 of 9 controlled experiments, with a median advance of 1,200 training steps. The leading margin grows monotonically as the training-set fraction decreases, suggesting SSR captures a structural precondition for generalization.

These findings extend Wang's Five Laws from a static diagnostic framework to a dynamic one: SSR not only characterizes well-trained models but also tracks the path toward generalization during training. The mechanistic hypothesis connecting SSR phase transitions to DFT basis convergence provides a bridge between the weight-space and activation-space perspectives on grokking.

Most importantly, the exact convergence of SSR to zero under extended training (SSR = 0.000000 at step 41,000) suggests that Wang's Second Law is not merely an empirical regularity but a **mathematical characterization of task structure**: the SSR lower bound measures how far a task's solution space is from having a unique spectral fixed point. Closed tasks with a unique optimal representation (modular arithmetic via the DFT basis) allow exact spectral alignment; open-ended tasks (natural language) do not. This distinction connects the static spectral geometry of large language models to the training dynamics of small algorithmic models, and elevates SSR from a diagnostic tool to a **fundamental quantity in the theory of learning**.

Broader Impact Statement

This work introduces a lightweight training monitoring signal (SSR) that could reduce compute costs associated with grokking detection. We do not foresee negative societal impacts.

References

Anonymous. Static weight diagnostics for LLM reasoning: Five universal spectral laws of transformer attention (Wang's Five Laws). *Transactions on Machine Learning Research*, 2026. Under review; withheld for anonymous review.

Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations*, 2023. URL `https://openreview.net/forum?id=9XFSbDZseq`.

Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022.

David Yunis, Tian Peng, Punit Bhatt, Jean Mercat, Liu Ziyin, and Preetum Nakkiran. Approaching deep learning through the spectral dynamics of weights. *Transactions on Machine Learning Research*, 2024. URL `https://openreview.net/forum?id=spectral-dynamics`.

A Code and Reproducibility

All training and analysis code is provided in the supplementary material (`grokking.zip`, four Python files <700 lines total, plus `README.txt` with step-by-step instructions).

File	Purpose	Output
`train_grokking.py`	Single 60K-step run (seed= 42, $\alpha = 0.4$)	`grokking_metrics.csv`
`train_grokking_v2.py`	9-experiment matrix (3 seeds $\times$ 3 α)	9 CSV files
`plot_ssr_grokking.py`	Reproduces Figure 3	`ssr_grokking_alignment.pdf`
`plot_ssr_grokking_v2.py`	Reproduces Figures 1, 2	`grokking_*.pdf`

All experiments run on a single consumer GPU (6 GB VRAM) in under 90 minutes total. No internet connection, proprietary data, or paid API access is required. `weight_decay=1.0` is critical for grokking and must not be changed.

B Catalog of Open Research Directions

SSR and the geometric framework of Wang's Five Laws (Anonymous, 2026) can be computed from static weights in milliseconds, with no inference, no labels, and no held-out data. Below we list research directions that become tractable once these tools are available. Each item is a question or a proposed method, not a validated result; we state them here to give the community a shared starting point.

A. Training Process Monitoring

1. **Early stopping without a validation set.** The SSR phase transition (peak then drop) is detectable from weights alone. Can it replace validation loss as the signal for when to stop or reduce the learning rate?

2. **Warmup end detection.** The first sustained SSR decrease may mark the natural end of the warmup phase.

3. **Stagnation detection.** When SSR stops falling, increasing batch size or perturbing the learning rate may help; SSR provides the trigger.

4. **Gradient health proxy.** A sudden SSR *increase* (rather than decrease) may signal gradient explosion or a corrupted batch before it shows in the loss.

5. **Multi-task balance.** In multi-task training, per-task SSR tracks which tasks are undertrained, enabling dynamic reweighting of the data mixture.

B. Checkpoint Management

1. **Cheap checkpoint ranking.** Among N saved checkpoints, rank by SSR and run full benchmarks only on the top K.

2. **Best checkpoint without benchmarks.** The checkpoint with the lowest deep-layer SSR is predicted to generalize best.

3. **Training dashboard.** An SSR curve alongside the loss curve gives a second, independent view of training health.

4. **Resume-training sanity check.** Computing SSR before resuming from a checkpoint can catch silent corruption.

C. Alignment Training

1. **RL improvement check.** After each RL round, does SSR improve? A model that scores higher on rewards but shows no SSR improvement may be reward-hacking.
2. **Catastrophic forgetting warning.** SSR rising during supervised fine-tuning suggests the pre-training spectral structure is being overwritten.
3. **Alignment cost measurement.** The SSR change from base to aligned model gives a concrete number for the structural cost of alignment.
4. **Fine-tuning method comparison.** DPO, PPO, GRPO, and SFT leave different SSR fingerprints; comparing them reveals how each method changes internal structure.
5. **Data quality signal.** After rejection-sampling fine-tuning, SSR change estimates data quality without running any downstream evaluation.

D. Quantization and Compression

1. **Quantization damage prediction.** SSR rise after INT8 or INT4 quantization predicts quality loss before any inference.
2. **Quantization method selection.** GPTQ, AWQ, and SmoothQuant leave different SSR footprints; pick the method that preserves SSR best.
3. **Precision planning from first principles.** Law III gives an analytic formula for the maximum safe depth under FP16 or BF16, with no tuning required.
4. **Pruning order.** Prune heads with the worst SSR first; keep heads that are already well-aligned.
5. **Distillation target.** Train the student not just to match logits but to match the teacher's SSR profile layer by layer.
6. **Sparsification safety.** If sparsification raises SSR beyond a threshold, treat the operation as unsafe.

E. Architecture Design

1. **Head count search.** How fast does SSR converge under different numbers of heads? Use SSR convergence speed as the search criterion instead of validation loss.
2. **Depth upper bound from Law III.** The analytic formula $L_{\max} = \min(L_{\text{info}}, L_{\text{quant}}, L_{\text{dyn}})$ sets a principled ceiling on model depth before building.
3. **GQA ratio.** Under pseudo-bulk aggregation, different KV-head counts produce different SSR; use this to find the best GQA ratio.
4. **Positional encoding comparison.** RoPE, ALiBi, and NoPE each distort the singular-value geometry differently; SSR measures the distortion directly.
5. **Attention variant equivalence.** Do FlashAttention and standard attention converge to the same SSR profile? If yes, the variants are geometrically equivalent; if no, they are not.

F. Inference and Deployment

1. **Weight integrity check.** Run SSR on a deployed model periodically. If SSR drifts, the weights have changed— no inference needed to detect this.
2. **Model fingerprint.** Each model's SSR profile is a compact, verifiable signature of its weights.

3. **Online learning drift detection.** In continual learning, rising SSR flags that new data is distorting the pretrained spectral structure.

4. **Federated learning filter.** Before aggregating client updates, check that each client's SSR profile is consistent with the global model.

5. **Provenance auditing.** SSR profiles can help trace which checkpoint a deployed model came from.

G. Evaluation and Benchmarking

1. **Inference-free model ranking.** Wang Score $(1 - \mathrm{SSR})$ ranks models without running a single prompt.

2. **Benchmark gaming detection.** A model that scores high on benchmarks but has poor SSR is likely overfitting to the benchmark, not genuinely reasoning.

3. **Pre-release quality gate.** Set a minimum SSR threshold before a model is cleared for release.

4. **Closed-source auditing.** A lab can publish its model's SSR report as a verifiable capability claim, without releasing the weights.

H. Spectral Fine-Tuning

Standard fine-tuning changes weights indirectly, through gradient descent on token predictions. Wang's Five Laws suggest a more direct route: edit the singular value decomposition of W_Q and W_K to push SSR toward zero. This family of methods bypasses gradient descent entirely for attention layers.

1. **Direct SSR minimization.** Solve $\min_{W_Q} \mathrm{SSR}(W_Q, W_K)$ subject to a norm constraint. No training data needed.

2. **Block-wise spectral editing.** Partition W_Q into spatial blocks and apply targeted singular value shifts within each block, leaving the rest intact.

3. **Rotation alignment.** Given fixed W_K, find a rotation $R \in \mathrm{SO}(d_h)$ such that RW_Q minimizes SSR. The solution is a geodesic on the Stiefel manifold—the shortest path between two orthogonal bases.

4. **Frequency-selective editing.** Large singular values may encode syntax; middle values, semantics; small values, fine-grained reasoning. Edit each frequency band separately rather than the whole matrix at once.

5. **K-to-Q derivation.** If $W_Q \approx f(W_K)$ for some structure-preserving map f, the query projection can be derived analytically from the key projection, skipping gradient descent altogether.

6. **SSR regularization.** Add $\lambda \sum_l \mathrm{SSR}_l$ to the fine-tuning loss to keep the spectral structure intact during SFT or RLHF.

7. **Cross-layer spectral transplant.** Copy the singular vectors of a well-aligned layer into a poorly-aligned layer as a warm-start initialization.

8. **Model merging by spectral interpolation.** Instead of averaging weights, interpolate the singular vectors of two models along the geodesic on the Stiefel manifold. This is geometrically cleaner than simple weight averaging.

I. Mixture-of-Experts

In a mixture-of-experts (MoE) model, each expert is a small network that handles a subset of inputs. Wang's Five Laws give a new lens for understanding what each expert has learned.

1. **Per-expert SSR profiling.** Compute SSR separately for each expert. Low-SSR experts are well-aligned and likely handle structured reasoning; high-SSR experts may be undertrained or specialized for noisy inputs.

2. **Expert specialization geometry.** Different experts may rotate their Q/K projections toward different subspaces— mathematics toward Fourier directions, code toward syntax-tree directions, language A toward one region of the embedding space and language B toward another. The angle between two experts' U_Q matrices measures how different their specializations are.

3. **SSR-guided routing.** Route tokens to the expert with the lowest SSR for the relevant task type, rather than using a learned router whose decisions are hard to interpret.

4. **Expert merging criterion.** Two experts whose SSR profiles are similar are good candidates for merging; two experts whose profiles differ should be kept separate.

5. **Expert splitting by spectral clustering.** An expert with high SSR and bimodal singular-value distribution may be trying to do two jobs at once. Split it into two experts by clustering its singular vectors—no extra training data required.

6. **Load balancing by spectral health.** Route more tokens to experts with low SSR (high spectral health) and fewer to experts with high SSR.

7. **Expert lifecycle management.** During training, monitor each expert's SSR; freeze experts that have converged and focus compute on those still improving.

8. **Cross-expert orthogonality.** Compute cosU between different experts' output subspaces. High overlap means the experts are redundant; low overlap means they are genuinely complementary.

J. Multilingual and Multimodal Models

1. **Language-specific spectral signature.** Do different languages activate different directions in V-space? If so, SSR gives a geometric map of how languages share or divide the embedding space.

2. **Cross-lingual transfer geometry.** How far does the U_Q subspace rotate when a model is fine-tuned from one language to another? This angle measures the structural cost of language transfer.

3. **Vision-language spectral gap.** Do vision tokens and text tokens produce the same SSR profile in a multimodal model? The data in Anonymous (2026) already show that vision encoders deviate from the text baseline on Law V—this deserves systematic study.

4. **Modality-specific fine-tuning.** Use per-modality SSR to decide which layers to fine-tune when adapting a multimodal model to a new modality.

5. **Cross-modal super-orthogonality.** Does cosU(Q,V) differ between vision and text attention layers? If vision layers are less super-orthogonal, retrieval and readout are less separated, which may explain cross-modal hallucination.

K. Safety and Alignment

1. **Jailbreak detection.** A successful jailbreak may leave a spectral fingerprint—SSR rising or cosU shifting in specific layers. This could be detected from weights alone, without looking at outputs.

2. **Backdoor detection.** A poisoned model may have anomalous SSR in the layers that implement the backdoor behavior.

3. **Alignment verification.** Does RLHF actually change the spectral structure of the reasoning layers? SSR gives a concrete, weight-based answer.
4. **Refusal mechanism geometry.** Which layers show unusual SSR when a model refuses a request? This could locate the refusal circuit without any activation analysis.
5. **Value geometry.** Do value-laden concepts occupy specific directions in V-space? If so, SSR could measure how much fine-tuning has shifted those directions.
6. **Alignment tax.** The SSR difference between a base model and its aligned version is a single number that captures the structural cost of making a model safer.

M. New Tools and Shared Infrastructure

1. **SpectralBench.** A public benchmark where models are ranked by SSR computed from their released weights. No prompts, no GPUs for inference, no risk of contamination.
2. **Wang Score leaderboard.** An open leaderboard based on $1 - \mathrm{SSR}$, updatable by anyone with access to the model weights.
3. **Spectral Diff.** A tool that shows, layer by layer, how SSR changes between two checkpoints— the weight-space equivalent of a code diff.
4. **Layer Health Report.** An automatic report of per-layer SSR, analogous to a static code analyzer but for neural network weights.
5. **Spectral Regression Test.** After each training run, check that SSR has not regressed relative to the previous checkpoint—the neural network equivalent of a software regression suite.
6. **Model genealogy by SSR distance.** Compute pairwise SSR distances between models to build a family tree of which models were fine-tuned from which base.
7. **Training trajectory atlas.** Visualize how SSR evolves during training for many different models and tasks, building a shared map of training dynamics in weight space.

L. Theoretical Connections

These are long-range conjectures. We state them as open questions, not claims.

1. **Renormalization group analogy.** SSR falls as depth increases, much like a renormalization group flow moves from short-scale (ultraviolet) to long-scale (infrared) degrees of freedom. Is there a precise correspondence?
2. **Universality class of the SSR transition.** The SSR phase transition during grokking looks like a second-order phase transition. Does it belong to a known universality class, such as the Ising model?
3. **Information geometry.** The Fisher information matrix and the singular value decomposition are both second-order structures on the parameter space. Is there a clean relationship between natural gradient descent and SSR minimization?
4. **Topological invariants.** The super-orthogonality of U-space (Law IV) may correspond to a topological invariant of the attention connection— a winding number or a Chern class. If so, the five laws have a topological interpretation that is robust to small perturbations of the weights.
5. **Fiber bundle formulation.** The Q/K/V weight matrices may define a fiber bundle over the representation space, with attention heads as local trivializations, the attention mechanism as a connection, SSR$\to$0 as the flatness condition, U-space super-orthogonality as the curvature between retrieval and readout fibers, and V-space random orthogonality as uniform fiber sampling. We offer this as a geometric picture, not a theorem. Its formal proof requires tools from differential geometry and operator algebra.

C Full Experiment Data

Table 2 reports SSR and Pearson r at the grokking step for all nine experiments.

Table 2: SSR and Pearson r at the grokking step t_{grok} for all nine experiments. Final SSR = value at step 20,000.

Experiment	t_{grok}	SSR at grok	r at grok	Final SSR
seed=0, frac=0.3	7700	0.000950	0.9997	0.000115
seed=0, frac=0.4	2800	0.009859	0.9892	0.000178
seed=0, frac=0.5	1400	0.025010	0.9935	0.000129
seed=42, frac=0.3	5200	0.000890	0.9999	0.000041
seed=42, frac=0.4	3000	0.006820	0.9959	0.000270
seed=42, frac=0.5	1300	0.020094	0.9785	0.000155
seed=1234, frac=0.3	5600	0.000983	0.9999	0.000088
seed=1234, frac=0.4	2600	0.010894	0.9921	0.000177
seed=1234, frac=0.5	2000	0.016192	0.9919	0.000360